

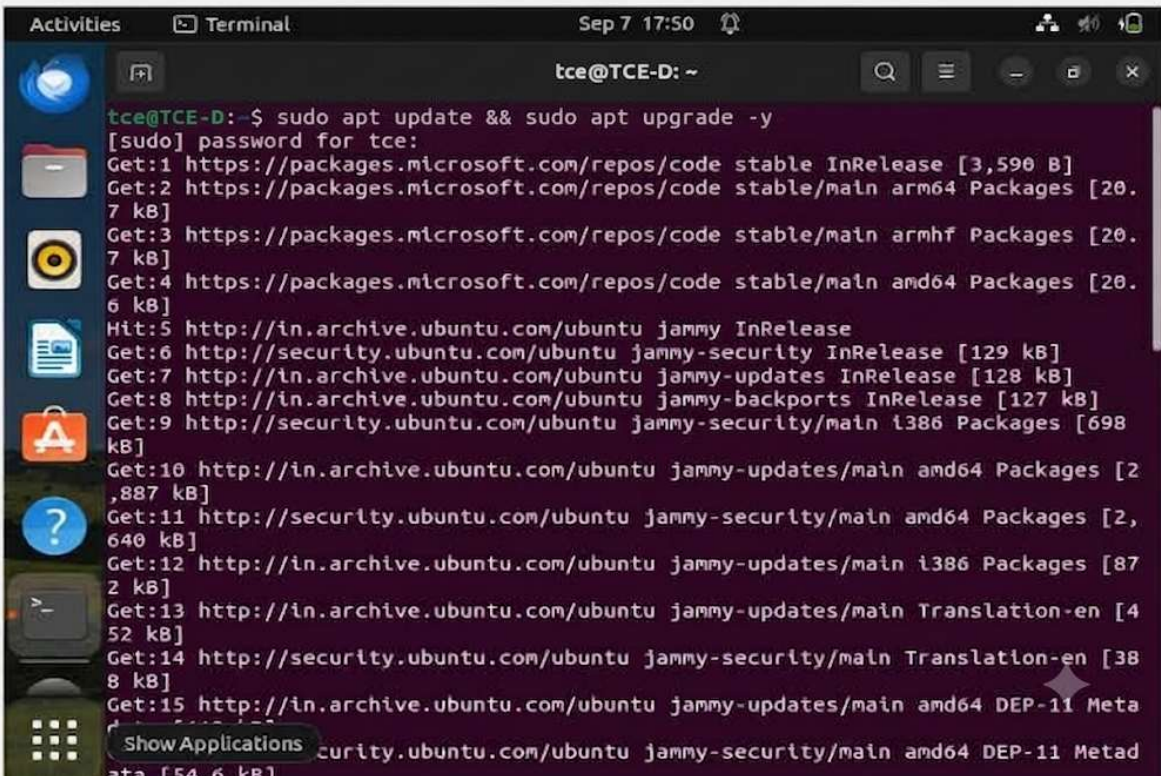
8) Installation of OpenStack and creation of Virtual machines

Aim:

To install of openstack and creation of virtual machine in linux system.

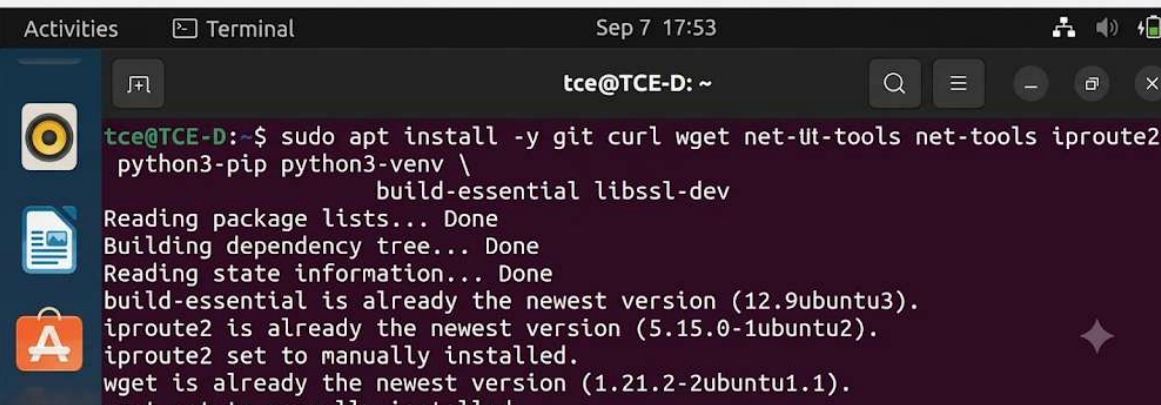
Steps:

1) Update all the file in the ubuntu by sudo apt update



```
tce@TCE-D: ~  
tce@TCE-D:~$ sudo apt update && sudo apt upgrade -y  
[sudo] password for tce:  
Get:1 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]  
Get:2 https://packages.microsoft.com/repos/code stable/main arm64 Packages [20.7 kB]  
Get:3 https://packages.microsoft.com/repos/code stable/main armhf Packages [20.7 kB]  
Get:4 https://packages.microsoft.com/repos/code stable/main amd64 Packages [20.6 kB]  
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy InRelease  
Get:6 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]  
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]  
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]  
Get:9 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [698 kB]  
Get:10 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2,887 kB]  
Get:11 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2,640 kB]  
Get:12 http://in.archive.ubuntu.com/ubuntu jammy-updates/main i386 Packages [872 kB]  
Get:13 http://in.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [452 kB]  
Get:14 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [388 kB]  
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 DEP-11 Meta  
data [54.6 kB]  
curity.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metad
```

2) Install the packages to build the openstack

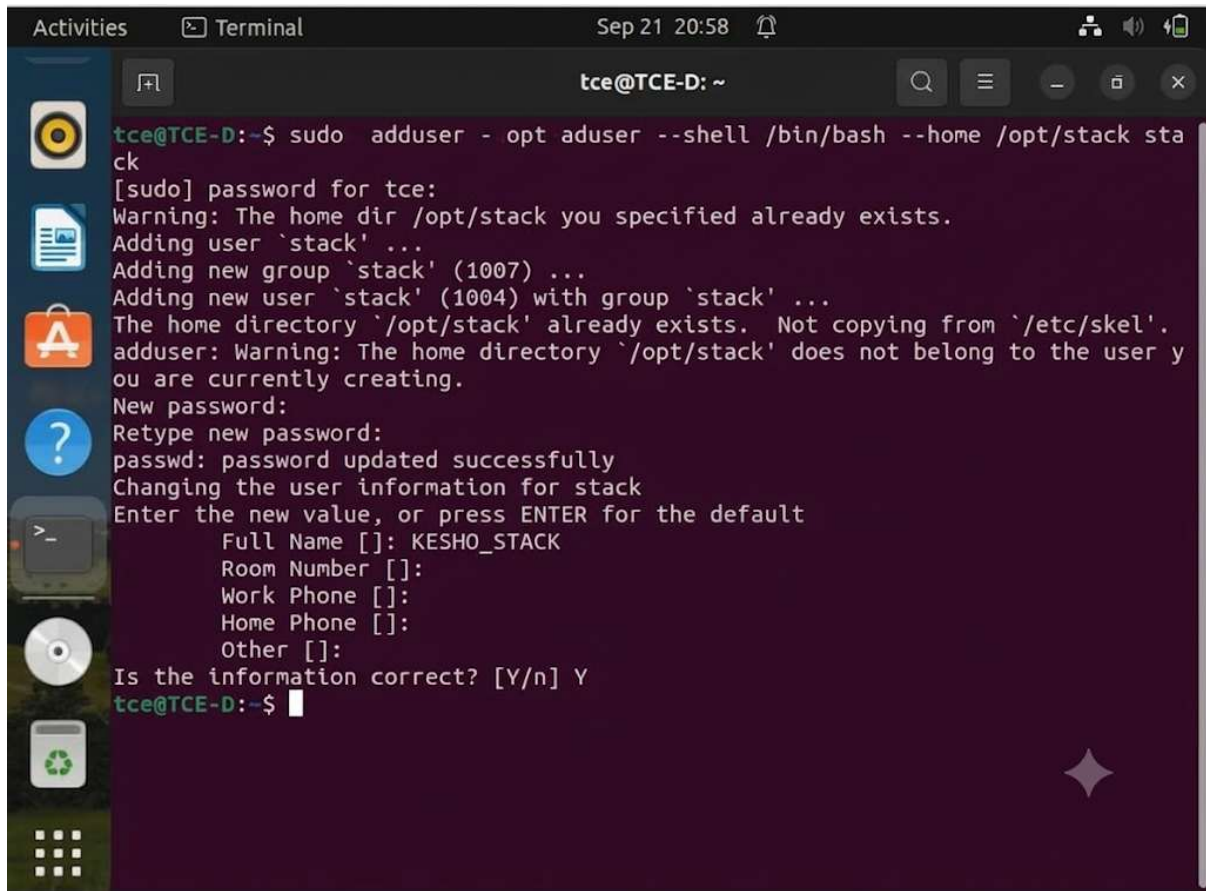


```
tce@TCE-D: ~  
tce@TCE-D:~$ sudo apt install -y git curl wget net-tl-tools net-tools iproute2  
python3-pip python3-venv \  
build-essential libssl-dev  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
build-essential is already the newest version (12.9ubuntu3).  
iproute2 is already the newest version (5.15.0-1ubuntu2).  
iproute2 set to manually installed.  
wget is already the newest version (1.21.2-2ubuntu1.1).  
wget set to manually installed.
```

DATE:

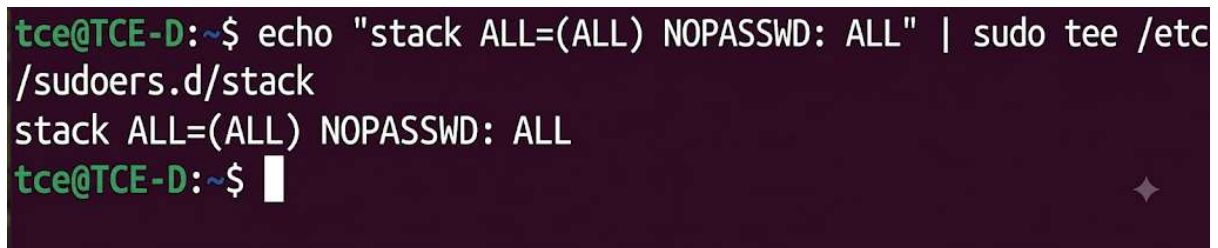
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3) Add the user stack to the ubuntu especially for the openstack and give the sudo privilege for the user stack and add to the group

A terminal window titled 'tce@TCE-D: ~' showing the execution of the 'sudo adduser' command to create a new user named 'stack'. The user is added with a home directory of '/opt/stack' and a shell of '/bin/bash'. The terminal output shows the password being set and user information being updated. The user 'stack' is added to the 'stack' group. The terminal output is as follows:

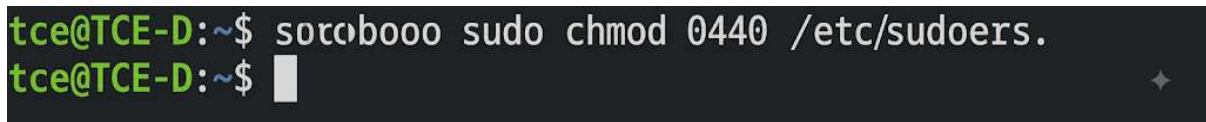
```
tce@TCE-D:~$ sudo adduser -o -s /bin/bash --home /opt/stack stack
[sudo] password for tce:
Warning: The home dir /opt/stack you specified already exists.
Adding user `stack' ...
Adding new group `stack' (1007) ...
Adding new user `stack' (1004) with group `stack' ...
The home directory `/opt/stack' already exists. Not copying from `/etc/skel'.
adduser: Warning: The home directory `/opt/stack' does not belong to the user y
ou are currently creating.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for stack
Enter the new value, or press ENTER for the default
  Full Name []: KESHO_STACK
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
tce@TCE-D:~$
```

4) Set no password for all

A terminal window showing the command 'echo "stack ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/stack' being executed. The output shows the command being run and the user 'stack' being added to the sudoers file with the NOPASSWD option. The terminal output is as follows:

```
tce@TCE-D:~$ echo "stack ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc
/sudoers.d/stack
stack ALL=(ALL) NOPASSWD: ALL
tce@TCE-D:~$
```

5) Give the privileges for the user stack

A terminal window showing the command 'sudo chmod 0440 /etc/sudoers.' being executed. The output shows the command being run and the permissions being set. The terminal output is as follows:

```
tce@TCE-D:~$ sudo chmod 0440 /etc/sudoers.
tce@TCE-D:~$
```

6) open the keshoo as sudo user by the command `sudo su - stack`

```
stack@TCE-D:~$ git clone https://git.openstack.org/openstack/devstack
Cloning into 'devstack'...
warning: redirecting to https://opendev.org/openstack/devstack/
remote: Enumerating objects: 52016, done.
remote: Counting objects: 100% (31604/31604), done.
remote: Compressing objects: 100% (10731/10731), done.
Receiving objects: 0% (292/52016), 108.00 KiB | 24.00 KiB/s
```

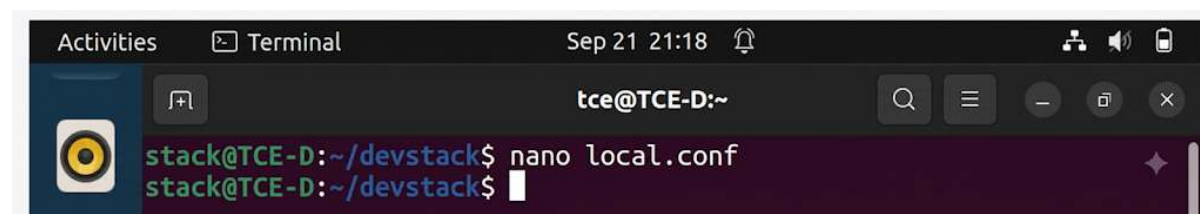
7) go to home directory and download the openstack from the git by cloning the git .

```
stack@TCE-D:~$ git clone https://git.openstack.org/openstack/devstack
Cloning into 'devstack'...
warning: redirecting to https://opendev.org/openstack/devstack/
remote: Enumerating objects: 52016, done.
remote: Counting objects: 100% (31604/31604), done.
remote: Compressing objects: 100% (10731/10731), done.
Receiving objects: 0% (292/52016), 108.00 KiB | 24.00 KiB/s
```

8) Move to the devstack directory

```
Resolving deltas: 100% (50502/50502), done.
tce@TCE-D:~$ cd devstack
tce@TCE-D:~$ cd
tce@TCE-D:~/devstack$
```

9) Change the configuration file in the local.conf

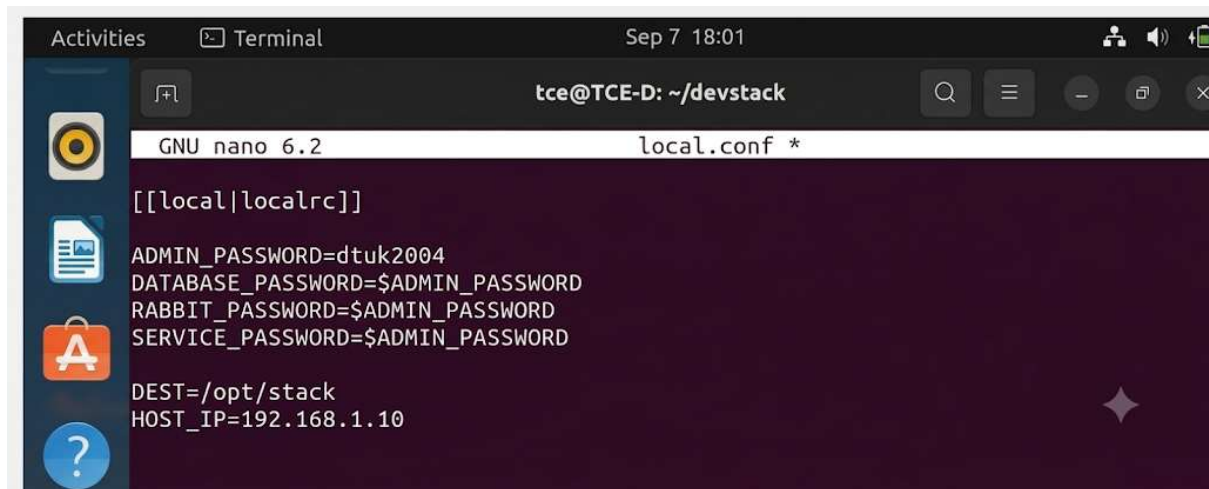


The screenshot shows a terminal window titled 'Terminal' with the date and time 'Sep 21 21:18'. The prompt is 'tce@TCE-D:~'. The user has navigated to the devstack directory and is now running the command 'nano local.conf'. The terminal output shows the prompt 'stack@TCE-D:~/devstack\$' followed by the command 'nano local.conf' and another prompt 'stack@TCE-D:~/devstack\$'.

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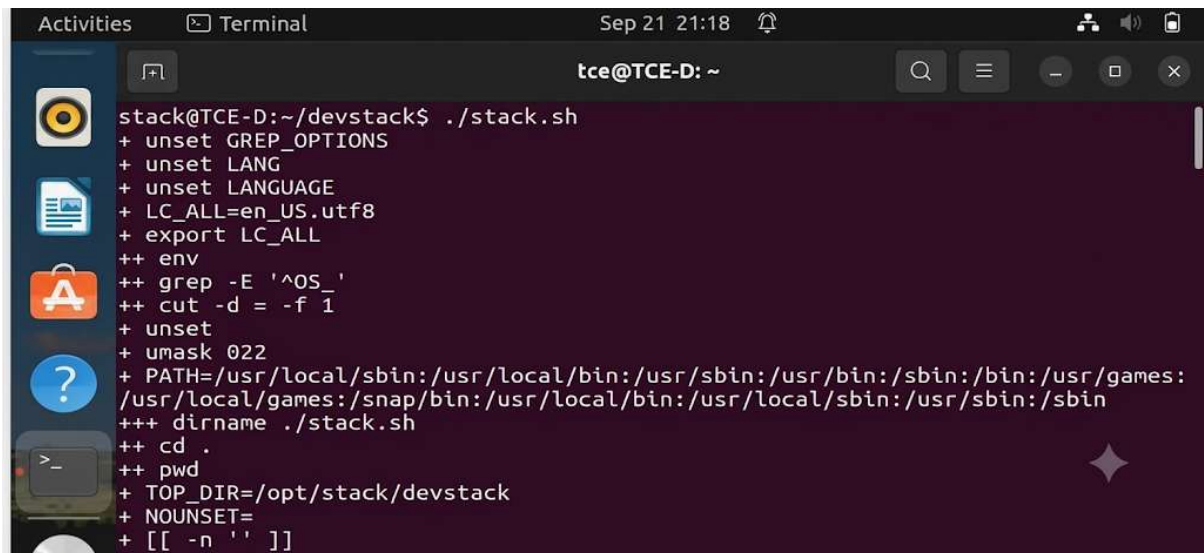
Update the below line and give password for each



A terminal window titled 'tce@TCE-D: ~/devstack' shows the GNU nano 6.2 editor editing a file named 'local.conf'. The file content is as follows:

```
[[local|localrc]]  
  
ADMIN_PASSWORD=dtuk2004  
DATABASE_PASSWORD=$ADMIN_PASSWORD  
RABBIT_PASSWORD=$ADMIN_PASSWORD  
SERVICE_PASSWORD=$ADMIN_PASSWORD  
  
DEST=/opt/stack  
HOST_IP=192.168.1.10
```

10) Run the openstack by the ./stack.sh

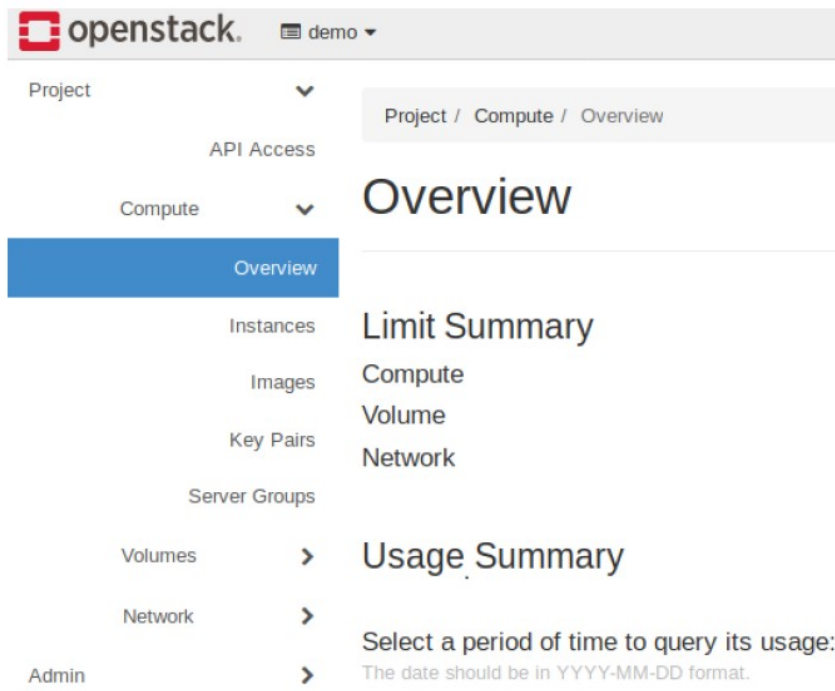


A terminal window titled 'tce@TCE-D: ~' shows the execution of the command './stack.sh'. The output is as follows:

```
stack@TCE-D:~/devstack$ ./stack.sh  
+ unset GREP_OPTIONS  
+ unset LANG  
+ unset LANGUAGE  
+ LC_ALL=en_US.utf8  
+ export LC_ALL  
++ env  
++ grep -E '^OS_'  
++ cut -d = -f 1  
+ unset  
+ umask 022  
+ PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/usr/local/bin:/usr/local/sbin:/usr/sbin:/sbin  
+++ dirname ./stack.sh  
++ cd .  
++ pwd  
+ TOP_DIR=/opt/stack/devstack  
+ NOUNSET=  
+ [[ -n '' ]]
```

11) Visit the IP address that you have configured in the local.conf and login with the password configured in the local.conf





12) Create a Virtual Machine using openstack glance service using the following command

a) Source OpenStack Credentials.

```
source devstack/openrc admin admin
```

(Replace admin with your actual project and username if different). Verify Image and Flavor.

b) Checkout the image list

```
openstack image list
```

```
openstack flavor list
```

Note the name or ID of the image and flavor you intend to use.

c) Create a Key Pair (if not already done):

```
openstack keypair create mykey > mykey.pem
```

```
chmod 600 mykey.pem
```

d) Create a Network

DevStack typically creates a default network. You can list existing networks:

`openstack network list`. If you need to create a new one, this involves creating a network, subnet, and router, and connecting them. Launch the Virtual Machine.

```
openstack server create --image <image_name_or_id> --flavor <flavor_name_or_id>
--nic net-id=<network_id> --key-name mykey my-first-vm
```

Replace `<image_name_or_id>`, `<flavor_name_or_id>`, and `<network_id>` with your chosen values. `my-first-vm` is the desired name for your VM. Verify VM Status and IP Address.

```
openstack server list
```

Look for your VM and its assigned IP address. Access the VM.

```
ssh -i mykey.pem cirros@<vm_public_ip>
```

Result:

Thus the installation of openstack and creation of virtual machine in linux system has been executed successfully and output has been verified.